LAB 4

Design of a dual band PIFA Antenna

Objective

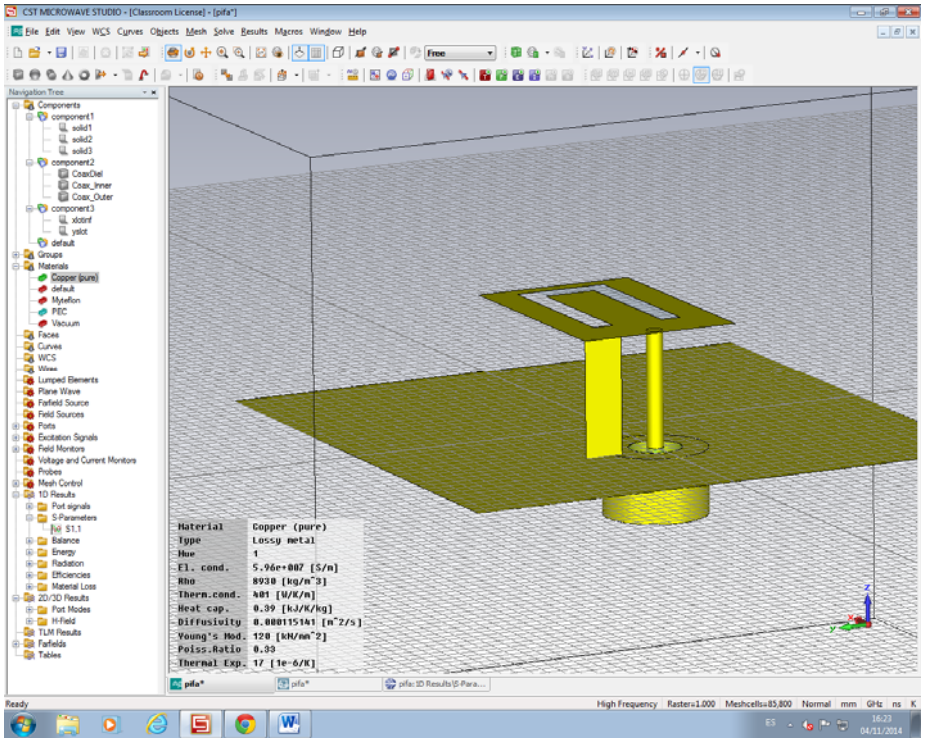
Once you have mastered the use of an antenna CAD software it is time you apply all your knowledge to the design of a real antenna. In this lab session you will have to design a dual band PIFA antenna to cover the 2.45 and 5 GHz WIFI band.

Pre- Work

- Identify the frequency bands that your antenna design must cover to support the WIFI standard at 2.45 and 5 GHz in Europe.
- Follow “P.Salonen et al.,Single-Feed Dual-Band Planar Inverted F Antenna with U-shaped Slot”, IEEE Transactions on Antennas and Propagation, vol. 48. No. 8, August 2000” to do a pre-design of the antenna to cover the specified frequency bands.

Lab-work

- Use CST studio to analyze your antenna design. Follow these guidelines
  READ THEM ALL BEFOR STARTING YOUR DESIGN:
  - Limit the ground plane size to 42 x 32 mm^2.
  - Limit the antenna height to 9 mm.
  - Implement the antenna short as a metal strip of 3 mm width as in the figure:



- You will use as a connector the Panel Mount Jack Receptacle removing the Teflon cover of the inner conductor after the ground plane. Use its dimensions.

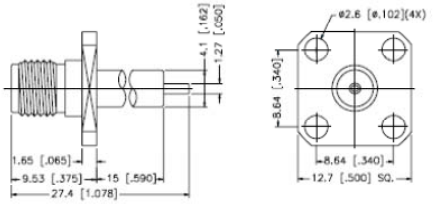
PANEL MOUNT JACK RECEPTACLE — 4-HOLE SQUARE FLANGE — EXPOSED TFE TYPE

P.N.	Cable Group	Finish	Insulation	Impedance	Crimp Tool
132146	N/A	Gold	Teflon	50	N/A
132302	N/A	Gold	Teflon	50	N/A

0.25 (.010") contact, 3.17 (.125") Teflon diameter

Remark: Extended teflon insulation available in various lengths. Contact Connex marketing department for part number and availability.





- In the solver settings set the accuracy to – 40 dB.
- Start with the antenna without the slot. Check the results and observe that you have a resonance around 2.45 GHz. At this moment do not readjust the antenna dimensions. Add the slot and observe that the second resonance appears.
- Follow the guidelines given in the paper and try to adjust the antenna response to the design objectives.
- For the final design specify the bandwidth and directivity.
- One final aspect is how the manufacturing tolerances will affect the antenna response. Introduce small variations in the antenna dimensions and identify the most critical dimensions and try to give manufacturing tolerance for them.

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Single-Feed Dual-Band Planar Inverted-F Antenna with U-Shaped Slot

Pekka Salonen, Mikko Keskilammi, and Markku Kivikoski

***Abstract*—The development of small integrated antennas plays a significant role in the progress of rapidly expanding wireless communication applications. This paper describes a novel dual-band planar inverted-F antenna (PIFA) for wireless local area network applications. The proposed PIFA uses single feed only. A novel top-plate geometry, a U-shaped slot, is discussed. An example is given for this novel slot shape for frequency bands of 2.4 and 5.2 GHz. Simulation based upon method of moments (MoM) is used to model the performance of the antenna. Comparisons with results measured on fabricated antenna structures are provided for simulations validation.**

***Index Terms*—Antennas, dual-band antennas, handset antennas, printed antennas, slot antennas, wireless communications.**

I. INTRODUCTION

The rapid progress in wireless communications promises to make interactive voice, data, and video services available anytime and anyplace. Wireless communication systems come in a variety of different sizes ranging from small hand-held devices to wireless local area networks. The integration of different radio modules into the same piece of equipment has created a need for dual-band antennas.

Good results for dual-band operation have been achieved with planar inverted-F antennas (PIFA) for portable handsets, reported in [1]–[4]. In [1] a capacitively coupled load and feed method was studied in order to reduce the size of the PIFA. The dual-band operation was a major concern in [2]–[4]. Dual-band operation was achieved with parasitic inverted-L elements in [2] and with etched slots in the radiating element in [3] and [4]. The slot in the radiating element gives a more compact design for the antenna and, thus, space–volume is saved.

In this paper, we describe a new dual-band internal antenna for wireless communication systems. The basic idea is to combine the geometries used in [2]–[4] with single feed only. A design example of a dual-band PIFA which has a new U-shaped slot is given for frequencies of 2.4 and 5.2 GHz. This antenna is fabricated and the measured results are compared to simulations.

II. ANTENNA GEOMETRY

A new configuration of dual-band PIFA, having a U-shaped slot on the radiating patch is presented. The geometry is shown in Fig. 1 for the U-shaped slot. The length l and width w of the PIFA determines the lower resonant frequency, which can be approximated by the formula [4]

$$f_{l0} = \frac{c}{4(w+l)} \tag{1}$$

where

c	velocity of light;
l and w	length and width of the radiating element;
f_{l0}	lower operating frequency.

The upper resonant frequency for a U-shaped slot PIFA can be determined approximately from (1) in which length l and width w are replaced by l_2 and w_2, respectively, as shown in Fig. 1. For the upper frequency band (1) gives a slightly lower frequency value compared to simulations and experiment.

This dual-band antenna has almost the same size as a single-band planar inverted-F antenna operating at the lower frequency band. The radiating element was grounded by a shorting strip at its corner and fed near the shorting strip using coaxial cable. The antenna impedance can be easily matched to 50 Ω by controlling the feed position from the shorting strip.

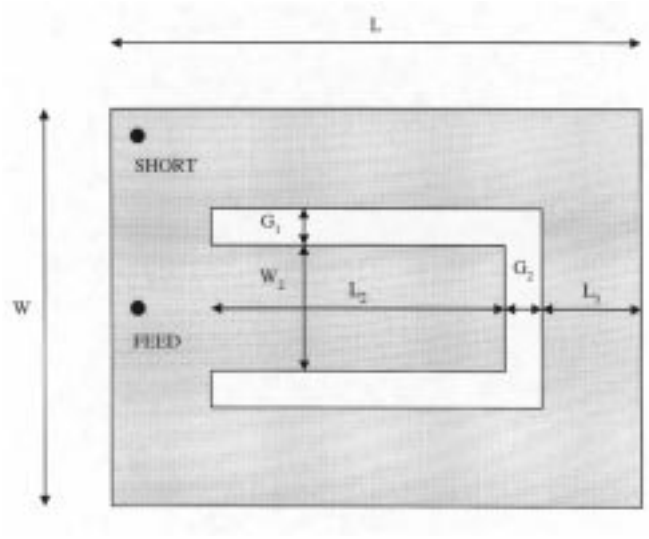


Fig. 1. Top-plate geometry configuration of single feed U-shaped slot dual-band PIFA.

TABLE I
TRENDS OF RESONANT FREQUENCY AND INPUT IMPEDANCE AS A FUNCTION OF THE GEOMETRICAL PARAMETERS OF THE U-SHAPED SLOT DUAL-BAND PIFA

	F1	F2	SWR1	SWR2
W, L ↑	↓			
L2 ↑		↓		
W2 ↑		↓		
G ↑		↓	↓	↓

III. PARAMETER STUDY

The sensitivity of resonant frequencies and input impedance to the geometry was studied for four parameters that were found to have the most critical influence. The values for the input impedance were specified as the frequency bandwidth in which the voltage standing wave ratio (VSWR) is less than 2 : 1. In this paper, the method of moments (MoM) has been used to allow modeling and simulation of the dual-band antenna.

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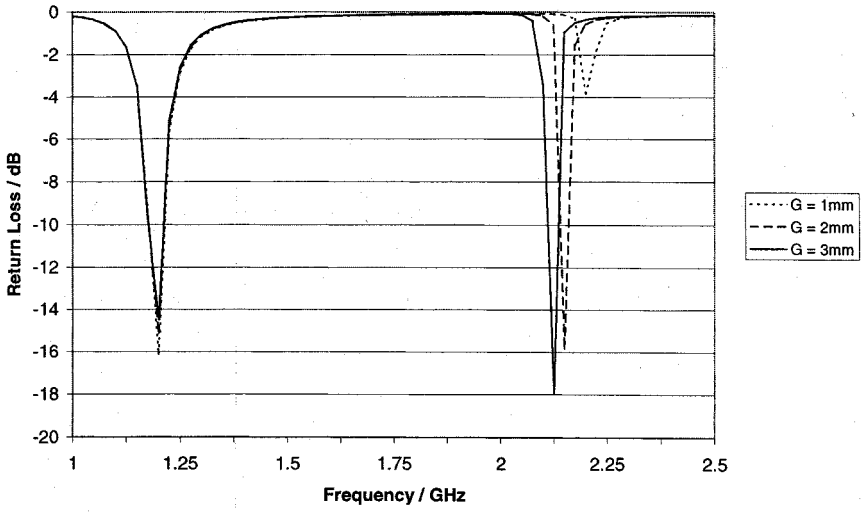


Fig. 2. Return loss of the U-shaped slot PIFA as a function of gap width in which $G1$ and $G2$ are equal length. Increase in $G1$ and $G2$ lowers the upper resonant frequency and null depth of the return loss. All other dimensions are fixed.

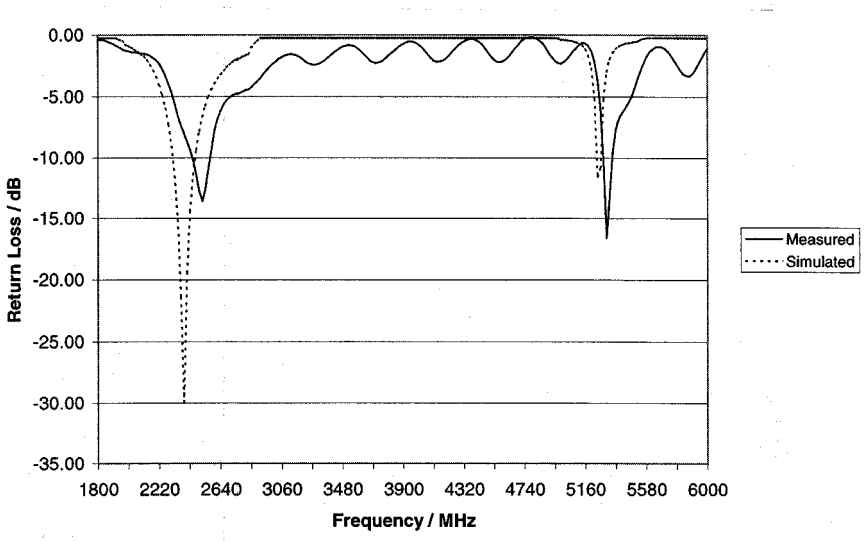


Fig. 3. Measured and simulated return loss.

The most critical parameters controlling the resonant frequencies of a dual-band U-shaped slot PIFA are summarized in Table I.

1) W—width of radiating element.
2) L—length of radiating element.
3) W_2—width of inner radiating element.
4) L_2—length of inner radiating element.

To understand the operation of our design we begin with a conventional PIFA (with the slot is removed) using air dielectric with dimensions $(l,\ w) = (40,\ 25)$ mm, height $h = 10$ mm in which a shorting strip of width $w_{\rm short} = 9$ mm is located at one end of the plate as shown in Fig. 1. The coax feed is connected directly to the top plate at the same edge as the shorting strip. The distance of the feed location in the middle of the top plate is also shown in Fig. 1. The resultant resonant frequency from the method of moments (MoM) simulations is 1.2 GHz, which agrees quite well with equation (1). To this conventional PIFA a U-shaped slot is added to determine its effects on the resonant frequencies and the bandwidth.

The dimensions of the slot are $L_2 = 27$ mm, $W_2 = 11$ mm, $L_3 = 8$ mm and $G = 2$ mm. The addition of the U-shaped slot had no effect on the lower resonant frequency and it remained at 1.2 GHz. The matching of the upper frequency can be controlled by the slot widths G_1 and G_2. The gap width G_2 is a compromise of input impedance because increase in G_2 tunes the lower frequency input impedance and mistunes the upper frequency input impedance, which as in Fig. 2. However, the best results in terms of input impedance matching are obtained when the gap widths G_1 and G_2 are equal.

This parameter study shows that for dual-band PIFA with U-shaped slot both frequencies can be determined independently. This makes the design procedure much simpler compared to dual-band PIFA with L-shaped slot.

IV. Experimental Validation

A dual-band antenna prototype was designed using U-shaped slot topology with the results in Section III for frequency bands 2.4 and 5.2 GHz. In order to characterize the antenna an HP8722D network analyzer was used to measure the input return loss of the antenna as a function of frequency.

The antenna was fabricated by etching the slot figure on a top plate, of 0.2-mm-thick copper and soldering it on a copper groundplane with a length of 180 mm and a width of 95 mm. The shorting strip dimensions were height 9 mm and width 3 mm. It should be emphasized here that the etching process was not accurate, which caused errors to the dimensions of the antenna. Thus, the 0.2-mm change in dimensions had the effect of altering the resonant frequencies. Also, the bandwidths of the antenna depend on the size and the shape of the groundplane, which is not optimized for these frequencies.

The dimensions for the U-shaped slot dual-band PIFA were $L = 15$ mm, $W = 12.5$ mm, $L_2 = 10.85$ mm, $W_2 = 4.2$ mm, G_1, $G_2 = 1.4$ mm and $L_3 = 1.25$ mm. The measured and simulated return loss of the U-shaped slot PIFA is shown in Fig. 3. The bandwidth of the lower resonant frequency is 230 MHz (9.6%), which agrees well with the simulations. However, the null depth is only half of that in the simulations because of the fabrication inaccuracies. The upper resonant frequency is slightly shifted up for the same reason. The null depth and the bandwidth 130 MHz (4%) agrees well with the simulations. This slot configuration is versatile for most wireless communication applications as for bandwidths. The lower frequency bandwidth is enough, e.g., for GSM or DCS, and the upper band is enough, e.g., for Bluetooth, which is not possible with L-shaped slot PIFA.

V. Conclusion

This letter focuses on the development of a dual-band planar inverted-F antenna using a novel top plate geometry. Series of simulations were used to investigate how different physical parts of the new U-shaped slot PIFA affect the input impedance, bandwidth, and resonant frequencies of the antenna. For the U-shaped slot PIFA both the lower and upper resonant frequency can be determined independently, which makes the design procedure simpler compared to L-shaped slot PIFA. The bandwidth of this antenna for the lower resonant frequency was 9.6% and for the upper resonant frequency bandwidth 4%. The wider bandwidth on the lower band of the U-shaped slot PIFA together with the easier design procedure makes it more versatile in wireless applications compared to L-shaped slot PIFA.

The results and design details on the antenna presented here can be used as starting points for engineers interested in utilizing low-profile dual-band planar inverted-F antennas in new wireless communications systems.

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Design of a Minimum-Loss Series-Fed Foldable Microstrip

C. G. Christodoulou, P. F. Wahid, M. Riad Mahbub, and M. C. Bailey

***Abstract*—The design and analysis of a series- fed, low-loss, inverted microstrip array antenna, operating at 1.413 GHz is presented. The array antenna is composed of two subarrays consisting of an equal number of microstrip patches connected together through a series microstrip line. The subarrays are coaxially fed 180° out of phase. This approach ensures a symmetric radiation pattern. The design approach for obtaining a low sidelobe ratio and a low loss microstrip array is accomplished using the IE3D code. Experimental and simulated data are presented and discussed.**

***Index Terms*—Microstrip arrays, spacecraft antennas.**

I. Introduction

One of the basic requirements of a radiometer antenna is to have very low radiation losses in order to maintain a desired radiometric resolution. In space applications, portability of the antenna is another requirement that must be satisfied. In this letter, these requirements are addressed through the design of a series-fed, foldable and inverted microstrip antenna array. This design minimizes the path length between the elements and, thus, minimizes ohmic losses [1], [2]. This reduction of losses makes the series configuration very attractive for radiometer applications. The inverted microstrip configuration is chosen because it provides less dispersion and dielectric losses than the conventional microstrip [3]. For the same characteristic impedance, substrate thickness, and for a comparable air gap, the attenuation due to conductor loss (usually the dominant loss mechanism) is improved by a factor of typically two to three [4]. The microstrip patches are supported by "*Rohacell*" foam material to reduce possible dielectric losses. The total antenna is composed of two subarray antennas, each consisting of an equal number of rectangular microstrip patch elements in series. The subarrays are fed 180° out of phase and an inverted ground plane is used to reduce radiation losses. This design yields low radiation losses and a radiation pattern with a narrow beamwidth and low sidelobes. A comparison between the measured and simulated data is presented and discussed.

II. Design Procedure

The first step is the determination of the dimensions of a single inverted microstrip patch. Next, additional patches are added to the array, one patch at a time to achieve the certain desired radiation pattern and input impedance. The inverted microstrip patch is the microstrip patch covered with dielectric layer and having the "*Rohacell*" foam as its dielectric substrate.

1) Calculation of the Patch Resonant Length: The resonant length of a single-patch microstrip antenna can be calculated from the empirical formula [5]

$$L = c \times (1 - 2V_R) \times (2 \cdot F_R \cdot \sqrt{\varepsilon_{eo}})^{-1} - 2 \cdot \Delta l \qquad (1)$$

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